

Exploring Image Segmentation Techniques to Detect High Latitude Dust on Satellite Images

1st Yichen Zhang

Lally School of Management
Rensselaer Polytechnic Institute
Troy, USA
yichenzhangycz@gmail.com

2nd Thilanka Munasinghe

Information Technology and Web Science
Rensselaer Polytechnic Institute
Troy, USA
munast@rpi.edu

Abstract—This paper presents a proposal to detect the High-Latitude Dust (HLD) events on the images captured by NASA satellites. We plan to use semantic segmentation techniques [1] to identify the HLD regions at the pixel level. The proposed model would distribute labels to specific pixels that we need to identify. After analyzing and learning those attributes, the trained model would identify the regions in new images with HLD events. However, in model training, the imbalanced data leads to a gap in the model's performance in coastal areas and sea areas, which means that even though the performance of dust detection over the ocean is good, the model cannot correctly detect the dust near or over the coast. Moreover, a gap exists between the different areas with different dust density. Thus, in this paper, we summarize the shortcomings of basic semantic segmentation models for HLD detection and propose a couple of solutions to improve such semantic segmentation models' performance.

I. INTRODUCTION

Over the last few decades, Moderate Resolution Imaging Spectroradiometer (MODIS) [2] on NASA's Terra satellite has observed dust events from cold, high latitudes. Some scientists think HLD could affect air quality, the reflectivity of snow, and even marine biology [3]; thus, it is an important phenomenon that we cannot ignore and requires adequate investigation.

A. High Latitude Dust

Particles that are less than 100 μm in diameter lifted from a surface and travel in the atmosphere by suspension is defined as "Dust" [3]. Moreover, high latitudes are defined as the areas $\geq 50^\circ\text{N}$ and $\geq 40^\circ\text{S}$ [3]. Thus, the High latitude dust (HLD) events we study in this work are the dust events that occur in high latitudes ($\geq 50^\circ\text{N}$ and $\geq 40^\circ\text{S}$). Some studies have confirmed eight essential high-latitude dust-event regions: Alaska, Canada, Eurasia, Greenland, Iceland, Antarctica, New Zealand, and Patagonia [4]. Take this phenomenon in Greenland as an example (Figure 1), where we could learn something more about the cause, influence, and importance of the HLD event. According to [5], glacier powder, i.e., fine particles composed of glaciers crumb and crushing rock, is the main dust ingredient found in Greenland. In the summer, the meltwater flow carries glacier flour or powder formed by glaciers to the valley and subsided in the floodplain near the

sea. Because of the decline of the water level in fall, the flood plain becomes dry, and the glacier dust becomes easier to be blown to the sky by the wind. Being able to detect such HLD events promptly by automatically analyzing big image data is a challenging task.



Fig. 1: An example of a HLD event due to glacier flour in Greenland Skies captured by the MODIS on NASA's Terra satellite and a sensor on the European Space Agency's Sentinel-2 (Image source: <https://earthobservatory.nasa.gov/images/92891/glacier-flour-in-greenland-skies>)

II. METHODOLOGY

In this section, we briefly summarize the work we undertook in developing the first trial model.

A. Data Description

The images were downloaded from Global Imagery Browse Services (GIBS) [6] as shapefiles, which contained the geographic information of left latitude, left longitude, right latitude, and right longitude of the area with HLD event. From the shapefile, we extracted the images with the geographic coordinates.

B. Image Labeling

We labeled images using CVAT [7], a tool that could be used for image labeling for deep learning algorithms. We

We express our gratitude to the RPI Information Technology and Web Science (ITWS) program for partially funding this research.

used a polygon to label the area with dust and downloaded the labeled information as a Common Objects in Content - Javascript Object Notation (COCO-JSON) file that stores the labeled information in digital form for model training.

C. Semantic Segmentation

The semantic segmentation technique we have selected for this research project is a common technique used in the geographical information system data analysis tasks [1]. DeeplabV3+ [1] is one of the advanced models in semantic segmentation, so we chose this library to perform our analysis. In the first trial, we labeled the pixels with dust and then put the labeled information into the model to learn the attributes of the dust pixels. With millions of dust pixels, the model would analyze and learn the attributes and recognize the pixel with dust in the test or validation images.

III. CONCLUSION AND FUTURE WORK

We have conducted a preliminary trial with a semantic segmentation algorithm to detect HLD events in the images captured by NASA's satellites.

However, when comparing the labeled original images and the predicted pixel areas, we find noticeable gaps. The first significant gap in the model performance exists in coastal areas and sea areas. While the performance of dust detection over the ocean is good, the model cannot correctly detect the dust near the coast. The models trained by other researchers also have this problem [8]. On images where most of the dust is over the sea, the pixel regions labeled with "dust" is enough for the model learning. However, when a small part of the dust background is a coastal area on the image, the HLD pixel region's size on the coastal area is not enough for the model training. The significant gap in pixel size leads to the gap in model performance between the two areas. As a solution to this problem, we are currently investigating the HLD detection as a multi semantic classification problem (i.e., dust in the coastal area and dust in the sea area as separate problems). This will yield better results compared to the first trial, in which we just used one classification (dust) to label the image data.

Moreover, another gap exists in the performance in areas with different densities. We notice that the relatively lower dust concentration areas could be predicted correctly, while the model cannot identify those with high dust concentration. We plan to train another multi semantic classification problem to address this problem (i.e., high-density dust and low-density dust). After comparing the two models, we will choose the best model.

REFERENCES

- [1] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation," in *ECCV*, 2018.
- [2] T. S. Pagano and R. M. Durham, "Moderate resolution imaging spectroradiometer (MODIS)," in *Sensor Systems for the Early Earth Observing System Platforms*, vol. 1939. International Society for Optics and Photonics, 1993, pp. 2–17.
- [3] J. E. Bullard, M. Baddock, T. Bradwell, J. Crusius, E. Darlington, D. Gaiero, S. Gasso, G. Gisladdottir, R. Hodgkins, R. McCulloch *et al.*, "High-latitude dust in the Earth system," *Reviews of Geophysics*, vol. 54, no. 2, pp. 447–485, 2016.
- [4] J. E. Bullard, "The distribution and biogeochemical importance of high-latitude dust in the Arctic and Southern Ocean-Antarctic regions," *Journal of Geophysical Research: Atmospheres*, vol. 122, no. 5, pp. 3098–3103, 2017.
- [5] A. Voiland. Glacier Flour in Greenland Skies. [Online]. Available: <https://earthobservatory.nasa.gov/images/92891/glacier-flour-in-greenland-skies>
- [6] M. Cechini, R. Boller, K. Baynes, J. Schmaltz, C. Thompson, J. Roberts, J. Rodriguez, B. King, M. Wong, Z. Rice *et al.*, "NASA GIBS and Worldview: Visualizing NASA's Earth Science Data for All to Explore," *NTRS - NASA Technical Reports Server*, 2018.
- [7] Powerful and efficient Computer Vision Annotation Tool (CVAT). [Online]. Available: <https://github.com/openvinotoolkit/cvat>
- [8] G. Priftis, B. Freitag, M. Ramasubramanian, I. Gurung, S. Gasso, M. Maskey, and R. Ramachandran, "Pixel Based Model For High Latitude Dust Detection," *NTRS - NASA Technical Reports Server*, 2019.